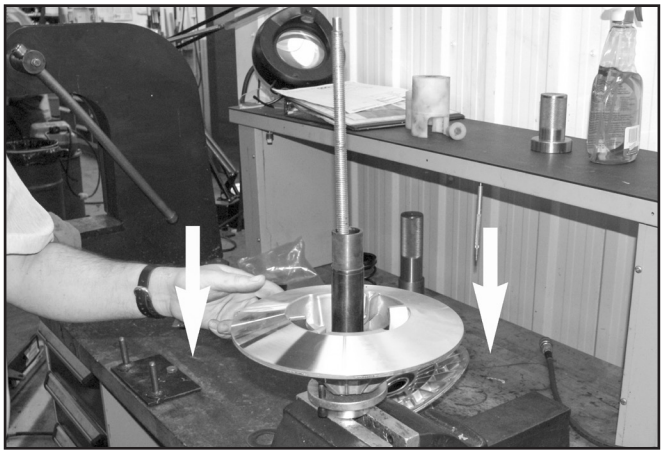
Assembling The Driven Clutch (Invance)

1. Place the Fixed Face to the Specialized Clutch Tool, 5055-0002, shaft facing up.
2. Assemble Shim Washer over shaft of fixed face.

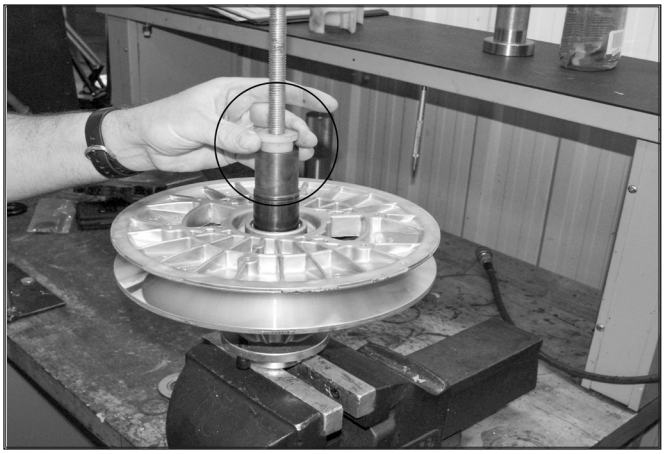



3. Slip the moveable sheave over the fixed shaft and let come to rest on the lower sheave.

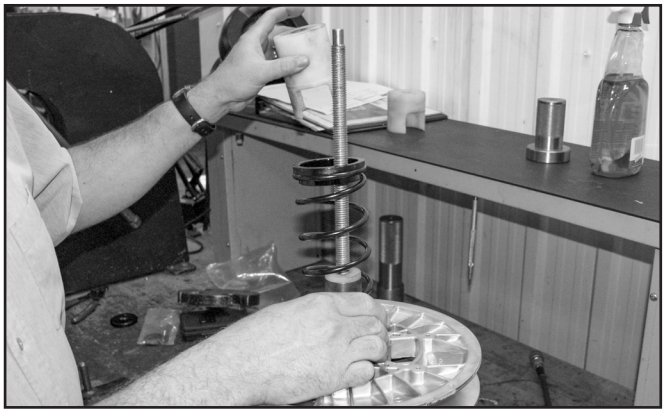
NOTE

Align the cam shoes with the slots provided in the lower fixed face.

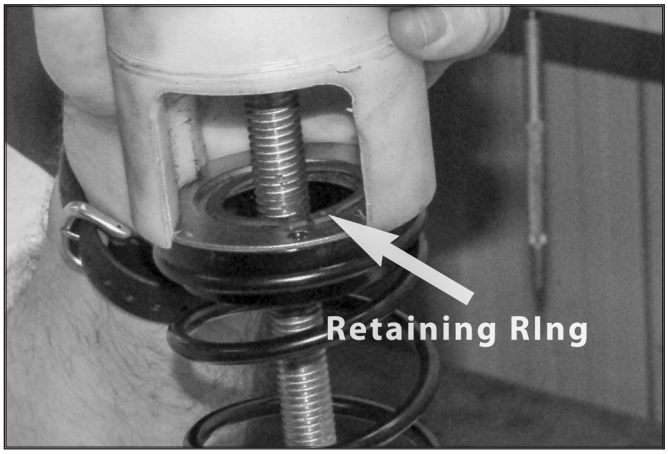
4. Place the nylon centering tool into the top of the fixed shaft.

5. Install clutch spring with plastic spring protector to the top.
6. Place steel washer to the top of plastic spring protector.

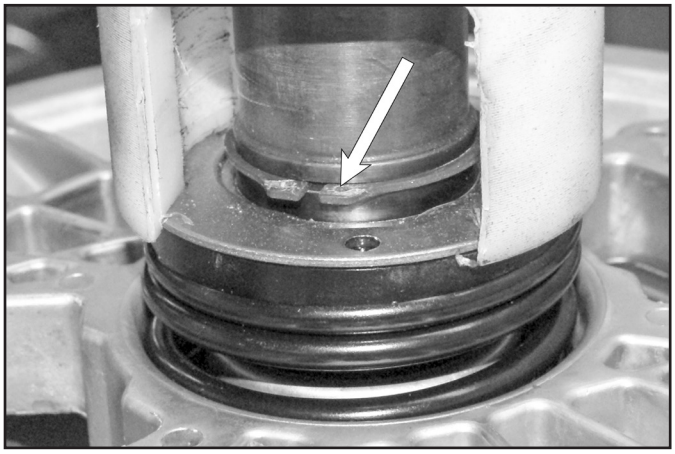



7. Place retaining ring into position over threaded rod before nylon assembly fork.
8. Compress the spring by turning the handle clockwise on the threaded rod.





9. Install retaining ring into ring groove of fixed pulley shaft.



Instance Torque Drive System Clutches (ITDS)

WARNING

Do not attempt to adjust, repair or replace the drive belt, clutches or any moving part while the engine is running. Doing so will cause injury. Before servicing the vehicle, disconnect the battery to prevent accidentally starting the engine.

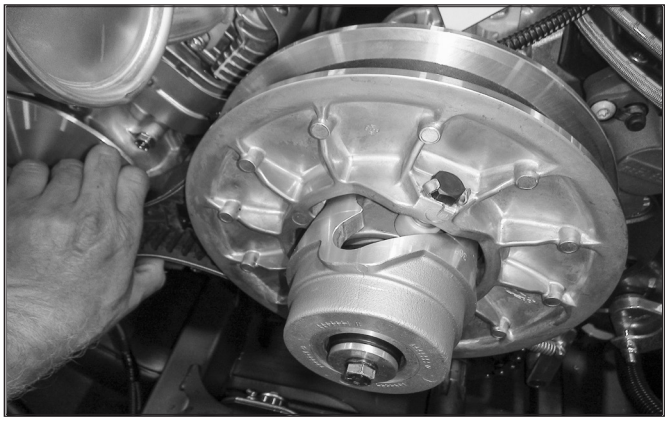
Keep the engine compartment hood, clutch guard (if applicable) and firewall securely in place when the engine is running. Severe injury can result if the drive belt, clutch components or other moving parts come loose.

If engine compartment inspection is necessary while the engine is running, use EXTREME CAUTION! Keep engine RPM low. Avoid standing directly in line with moving components. Use a mirror to view the components.

The drive belt transmits power from the driver clutch (on the engine) to the driven clutch (on the transmission). These components are located on the left side of the engine compartment.

Removing The Drive Belt (ITDS Clutch System)

1. Perform the servicing procedure, **Drive Belt Inspection.**
2. Squeeze the drive belt to spread driven clutch sheaves. Thread belt off driven clutch.

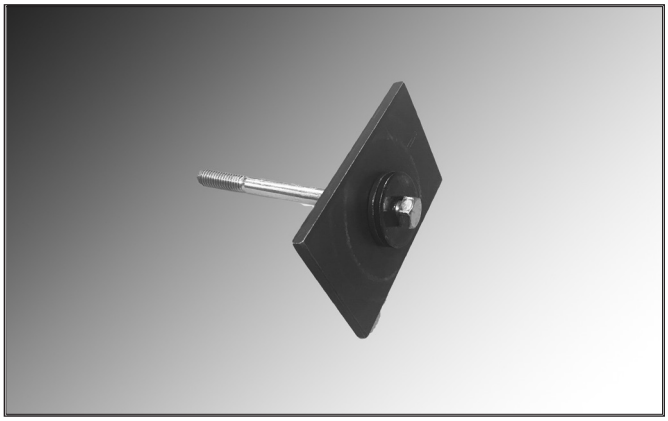
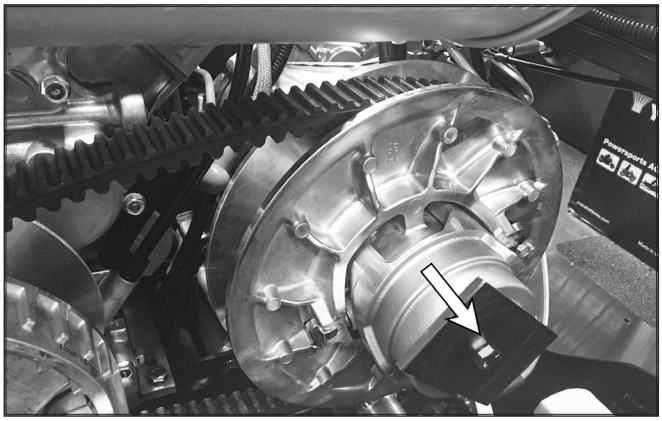



Installing The Drive Belt (ITDS Clutch System)

CAUTION

This procedure requires as specialized tool to spread the driven clutch sheaves.

1. Remove the bolt securing the driven clutch to the input shaft and insert it through a piece of flat bar stock (may or may not require some flat washers).
2. Thread the bolt into the driven clutch until the clutch sheaves spread far enough to easily remove the drive belt.

3. To install the drive belt, position the belt around the driver clutch first.
4. Ease the belt over the edge of the fixed face on the driven clutch and carefully thread the belt over the top while turning both driver and driven clutch clockwise.
5. Remove the clutch tool and reinstall the mounting bolt.

NOTE

Drive Belt alignment and tension are pre-set and are not adjustable.

Driver Clutch Removal (ITDS)

1. Remove the engine access cover.
2. Perform the servicing procedure, **Removing the Firewall**.
3. Perform the servicing procedure, **Removing the Drive Belt (ITDS Clutch System)**
4. Remove the fastener securing the Driver Clutch to the PTO and slide the clutch from the shaft.

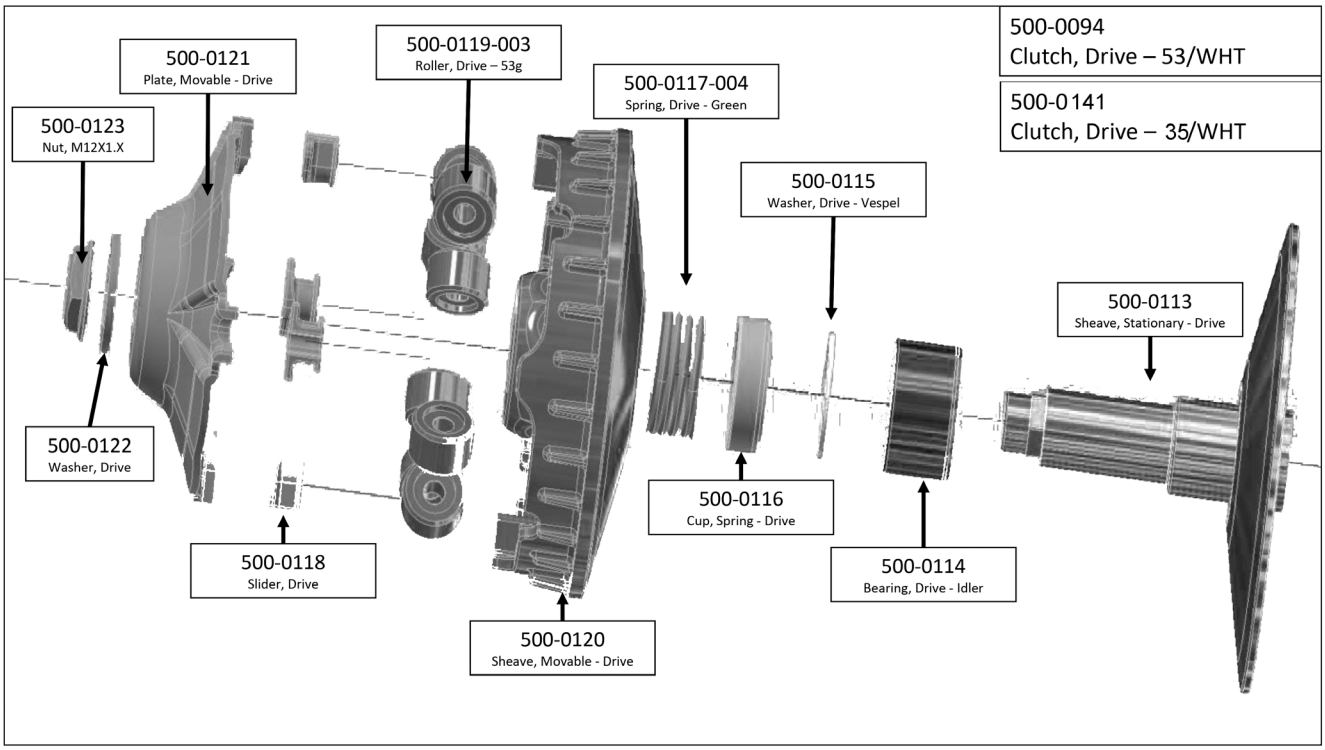
NOTE

The driver clutch must be slid from the PTO with the mounting bolt hardware within the clutch. The mounting bolt is too long to remove first.

The driver clutch may contact the lower body first before completely being free of the PTO, however the lower body will flex enough in that area to free the clutch.

5. Remove the clutch to a clean work area for further disassembly.

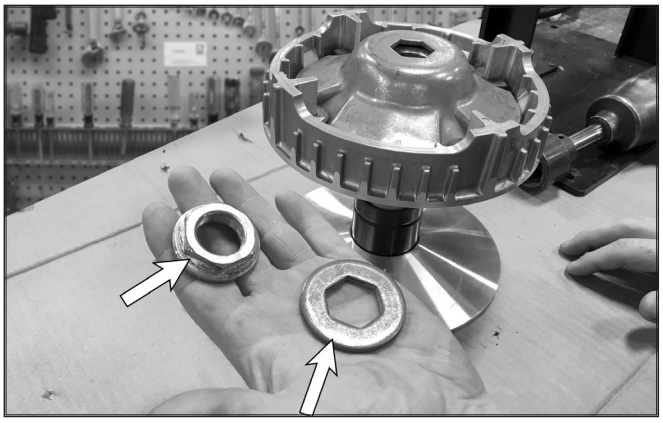
Driver Clutch Disassembly (ITDS)



1. Using a 1 5/16" impact socket, remove the 500-0123 Nut and 500-0122 Washer from clutch. *See following page.*

CAUTION

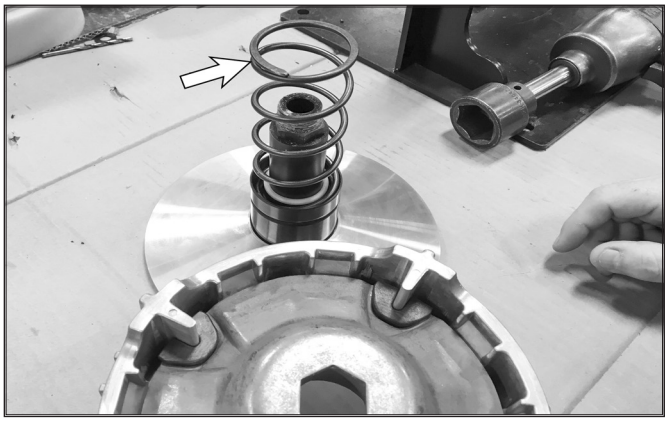
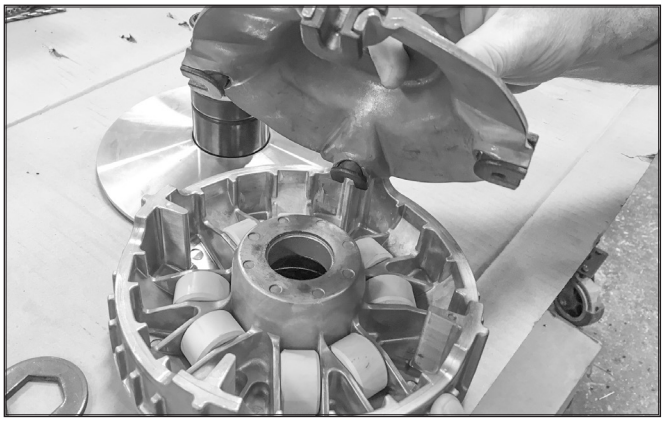
Clutch is under spring tension. Install a specialized clutch compressor tool before attempting to remove the nut and washer. Proceed with care slowly releasing the spring tension by backing off the compression tool once the nut and spacer have been removed.

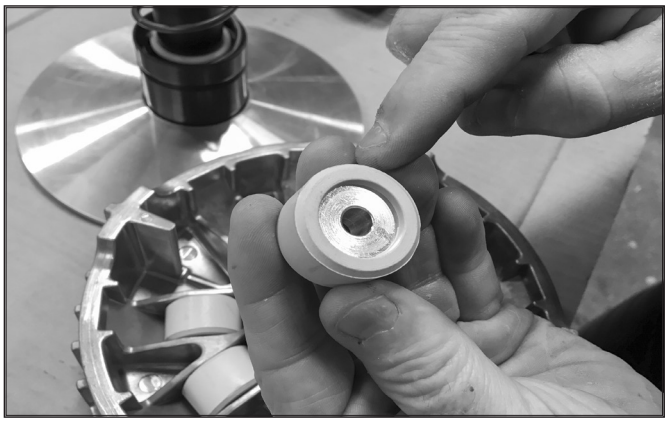
2. Lift off the moveable face to expose 500-0117-004 Spring, Drive and remove spring.
3. Remove 500-0121 Plate Moveable-Drive to expose 500-0119-003 Roller, Drive weights.

⚠ IMPORTANT

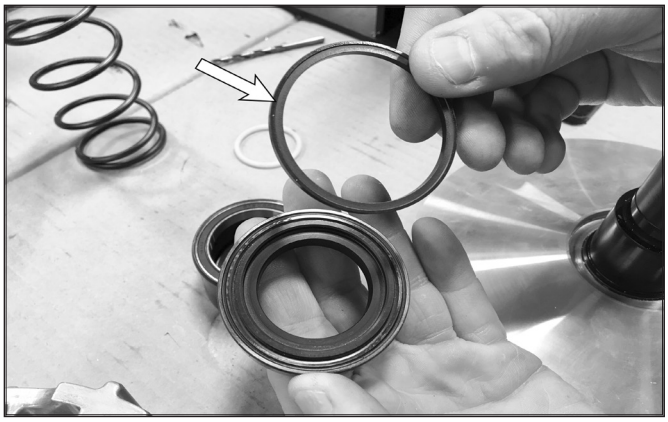
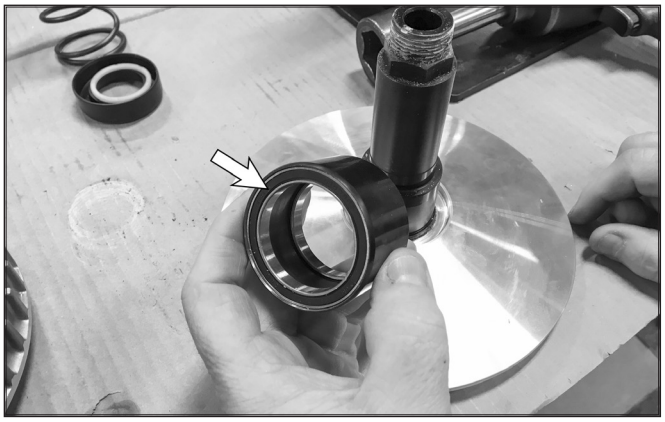
All 600/650/700/850/950 ARGO models utilize 53g weights. 800 ARGO Models utilize 35g weights. This can be distinguished by the larger hole in the 800 model clutch rollers.

4. Check for any damaged rollers.
5. Slip off 500-0116 Cup, Spring Drive and Nylon Ring.

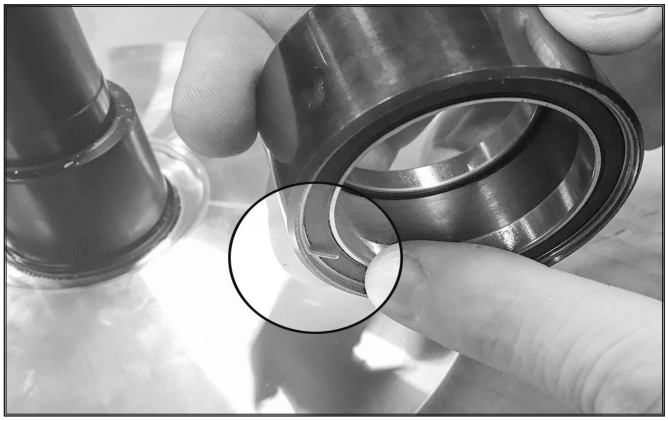



6. Locate 500-0115 Vespel Washer installed to the other side of the 500-0116 Cup, Spring Drive.
7. Slip 500-0114 Bearing, Drive-Idler from the fixed shaft.

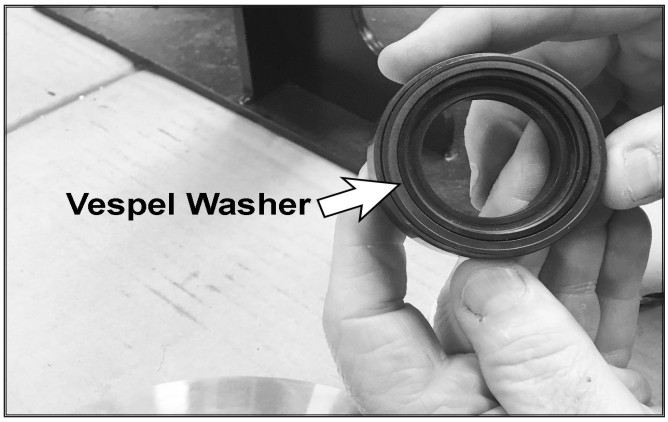



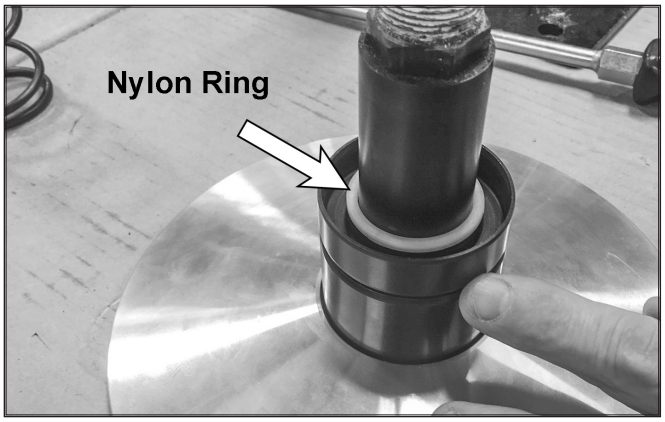
Driver Clutch Reassembly (ITDS)

1. Slip the 500-0114 Bearing, Drive-Idler on to the shaft of the fixed face. Orient the bearing with the spring clip facing down.

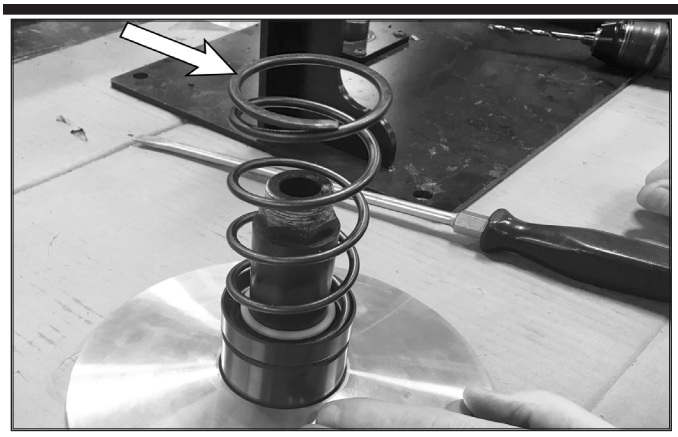
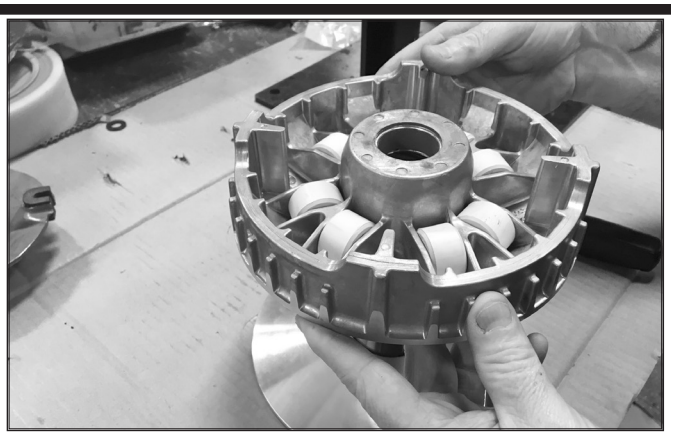



2. Place the 500-0115 Vespel Washer to the *bottom* of the 500-0116 Cup, Spring Drive and slip over the shaft followed by the Nylon Ring.

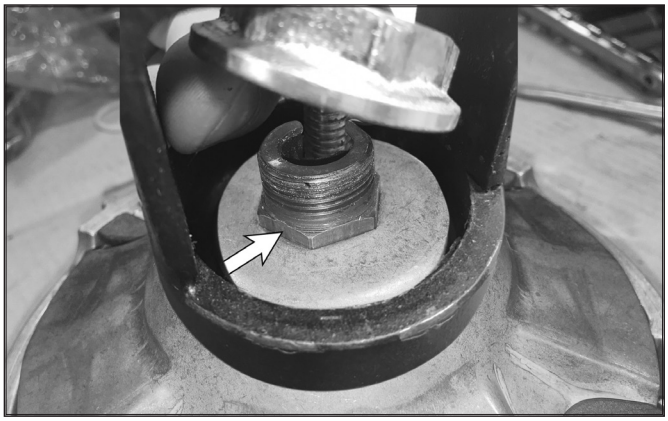




3. Install the 500-0117-004 Spring, Drive to the fixed shaft.
4. Install the 500-0120 Sheave, Moveable Drive with rollers to the fixed shaft.

5. Place 500-0121 Plate, Moveable Drive to the top of the 500-120 Sheave Moveable Drive. Align the slots in the upper cover with the cast aluminum locator pins in the sheave below.
6. Compress the spring using the compression tool until the hex of the fixed face shaft is visible.




7. Install 500-0122 Washer and 500-0123 Nut to clutch.
   Torque to 80 FT.LBS. (110 Nm).

Install The Driver Clutch (ITDS)

1. Place the clutch key into the keyway on the engine PTO.
2. Apply anti-seize compound to the output shaft.
3. Apply 242 Blue LOCTITE to the threads of the mounting bolt and insert it along with the washer hardware into the clutch first before sliding it onto the PTO shaft.
4. Align the keyway in the clutch with the key installed to the PTO shaft.
5. Slide the clutch onto the shaft and up to the shoulder of the PTO shaft.

CAUTION

The clutch should slide on by hand. Do NOT use force or tap on with a hammer.

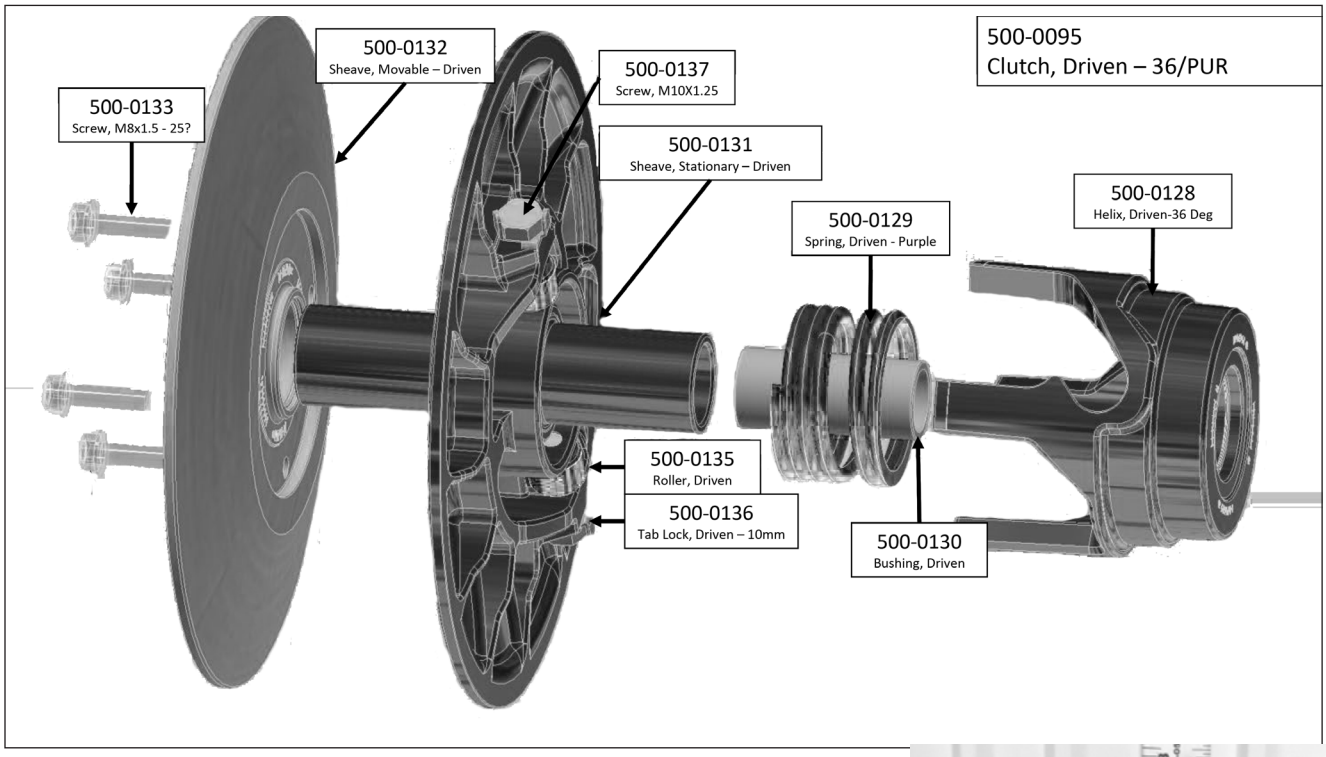
6. **Torque driver clutch bolt to 40 FT.LBS.**
7. Perform the servicing procedure, **Installing the Drive Belt (ITDS Clutch System).**

Removing The Driven Clutch (ITDS)

1. Perform the servicing procedure, **Removing The Firewall.**
2. Perform the servicing procedure, **Removing The Drive Belt (ITDS Clutch System).**
3. Locate the HHCS 3/8-16x3.5 Gr5 Zn, Lock Washer and Flat Washer securing the clutch to the input shaft and remove with 9/16" socket wrench.
4. The driven Clutch is installed to a straight keyed shaft. Slide the driven clutch off the input shaft.



Driven Clutch (ITDS)



Driven Clutch Disassembly (ITDS)

1. Perform the servicing procedure, **Removing The Firewall.**
2. Perform the servicing procedure, **Removing The Drive Belt (ITDS Clutch System).**
3. Perform the servicing procedure, R**emoving The Driven Clutch (ITDS)**
2. Use appropriate driven clutch compression tool to disassemble the driven clutch.

NOTE

This type of tool can be sourced online.

3. Install driven clutch on tool. Select proper spacer, cap and retainer that fits over the driven clutch post and puts pressure on the stationary sheave. Tighten securely.

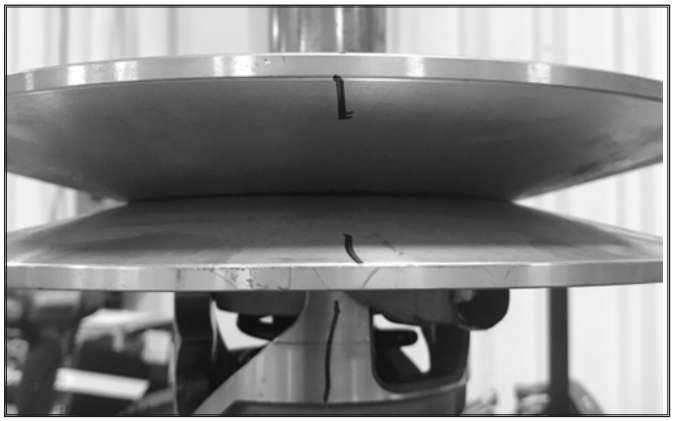


4. With a permanent marker, draw a straight line on the movable sheave, stationary sheave, and helix to note proper alignment when reassembling the driven clutch components.

NOTE

The sheaves are balanced components, so it is necessary to reinstall them properly.

5. Apply heat to screws to loosen the LOCTITE. Remove the four screws securing the stationary sheave to the helix.





6. Loosen the retainer on the tool noting that the driven clutch components are slowly separating.

CAUTION

If the sheaves and helix are not separating, tighten the tool retainer and inspect.

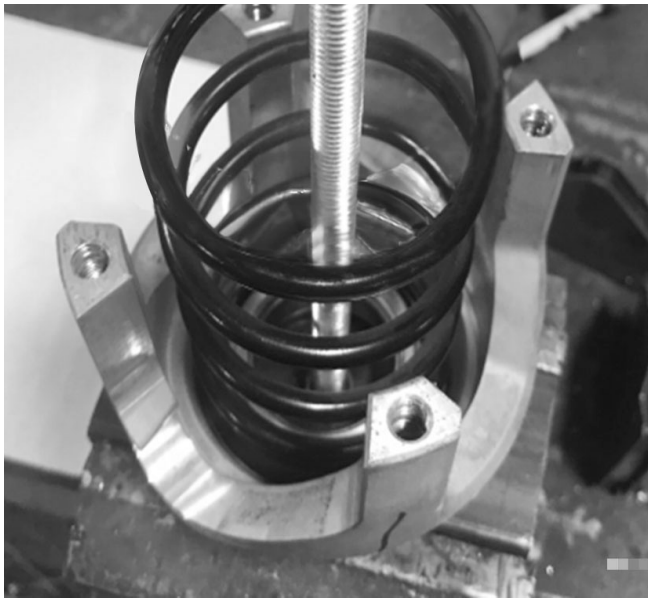
7. Remove the sheaves in assembly.
8. Remove spring.





Driven Clutch Reassembly (ITDS)

9. Install spring back into helix.
10. Install sheave assembly making sure to align the permanent marker lines from step 4.
11. Tighten retainer nut on tool slowly while aligning the helix with the sheaves.



⚠ NOTE

Make sure to the roller is placed on the correct side of the helix tower as seen in picture.



